HorizonTechX_Unemployment Analysis .i...

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RAM
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[3]
15s

```
from google.colab import files
uploaded = files.upload()
```

Choose Files Unemploym...11_2020.csv

Unemployment_Rate_upto_11_2020.csv(text/csv) - 18274 bytes, last modified: 11/5/2020 - 100% done

```
Saving Unemployment_Rate_upto_11_2020.csv to Unemployment_Rate_upto_11_2
```

[4]
0s

```
import pandas as pd

df = pd.read_csv("Unemployment_Rate_upto_11_2020.csv")
df.head()
```

	Region	Date	Frequency	Estimated Unemployment Rate (%)	Estimated Employed	Estimated Labour Participation Rate (%)	Regior
0	Andhra Pradesh	31-01-2020	M	5.48	16635535	41.02	Sc
1	Andhra Pradesh	29-02-2020	M	5.83	16545652	40.90	Sc
2	Andhra Pradesh	31-03-2020	M	5.79	15881197	39.18	Sc
3	Andhra Pradesh	30-04-2020	M	20.51	11336911	33.10	Sc
4	Andhra Pradesh	31-05-2020	M	17.43	12988845	36.46	Sc

Next steps: Generate code with df New interactive sheet

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[5]
0s

```
print(df.isnull().sum())
```

```
Region                                      0
 Date                                       0
 Frequency                                  0
 Estimated Unemployment Rate (%)            0
 Estimated Employed                         0
 Estimated Labour Participation Rate (%)    0
Region.1                                    0
longitude                                   0
latitude                                    0
dtype: int64
```

[6]
0s

```
df = df.dropna()
print(df.isnull().sum())
```

```
Region                                      0
 Date                                       0
 Frequency                                  0
 Estimated Unemployment Rate (%)            0
 Estimated Employed                         0
 Estimated Labour Participation Rate (%)    0
Region.1                                    0
longitude                                   0
latitude                                    0
dtype: int64
```

[7]
0s

```
print(df.head())
```

```
           Region         Date  Frequency   Estimated Unemployment Rate
0  Andhra Pradesh   31-01-2020          M                             5
1  Andhra Pradesh   29-02-2020          M                             5
2  Andhra Pradesh   31-03-2020          M                             5
3  Andhra Pradesh   30-04-2020          M                            20
4  Andhra Pradesh   31-05-2020          M                            17

    Estimated Employed   Estimated Labour Participation Rate (%) Region.
0             16635535                                     41.02    Sout
1             16545652                                     40.90    Sout
2             15881197                                     39.18    Sout
3             11336911                                     33.10    Sout
4             12988845                                     36.46    Sout

   longitude  latitude
0    15.9129     79.74
1    15.9129     79.74
2    15.9129     79.74
3    15.9129     79.74
4    15.9129     79.74
```

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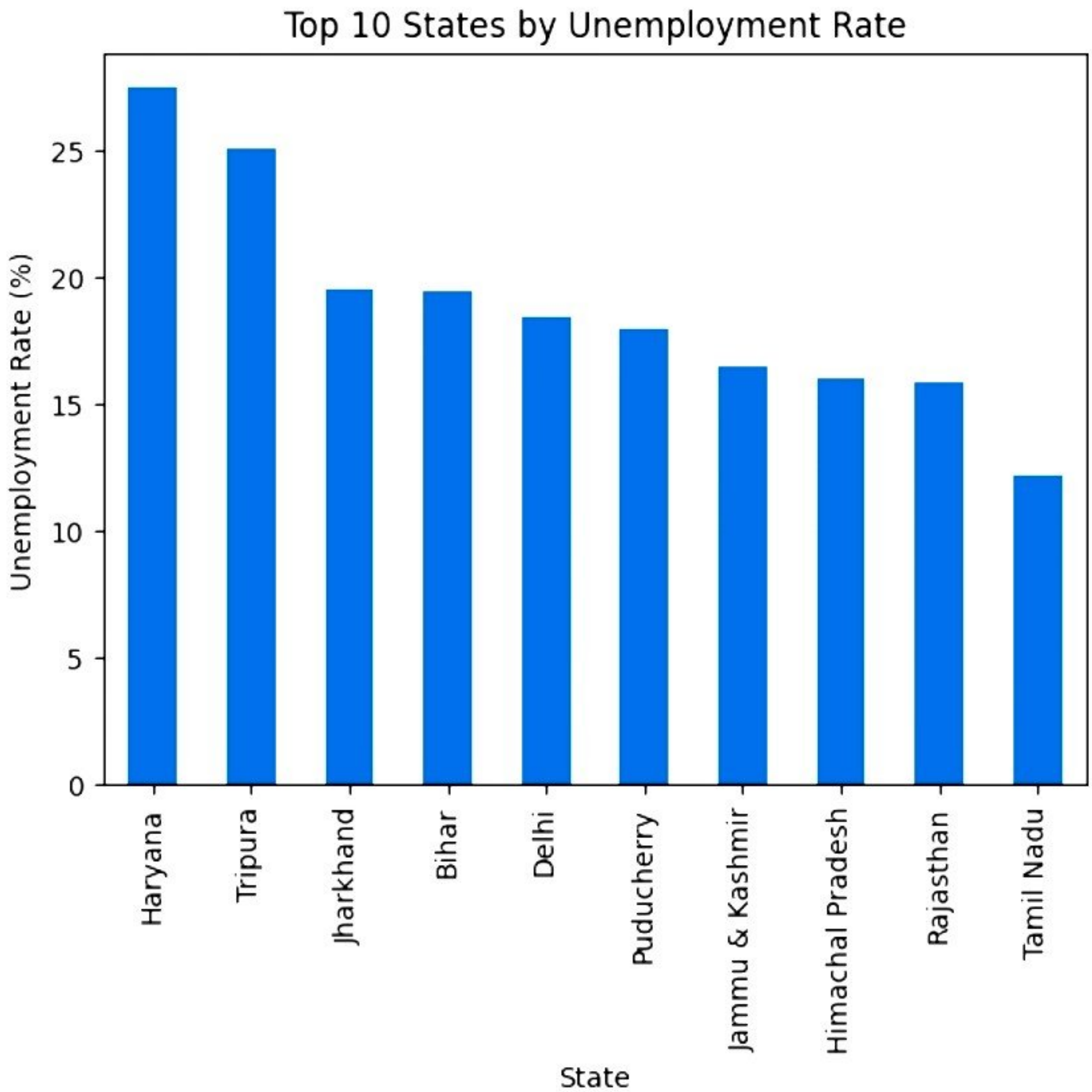
[8]
0s

```
import matplotlib.pyplot as plt

state_unemployment = df.groupby('Region')[' Estimated Unemployment Rate

state_unemployment.sort_values(ascending=False).head(10).plot(kind='bar

plt.title("Top 10 States by Unemployment Rate")
plt.xlabel("State")
plt.ylabel("Unemployment Rate (%)")
plt.show()
```



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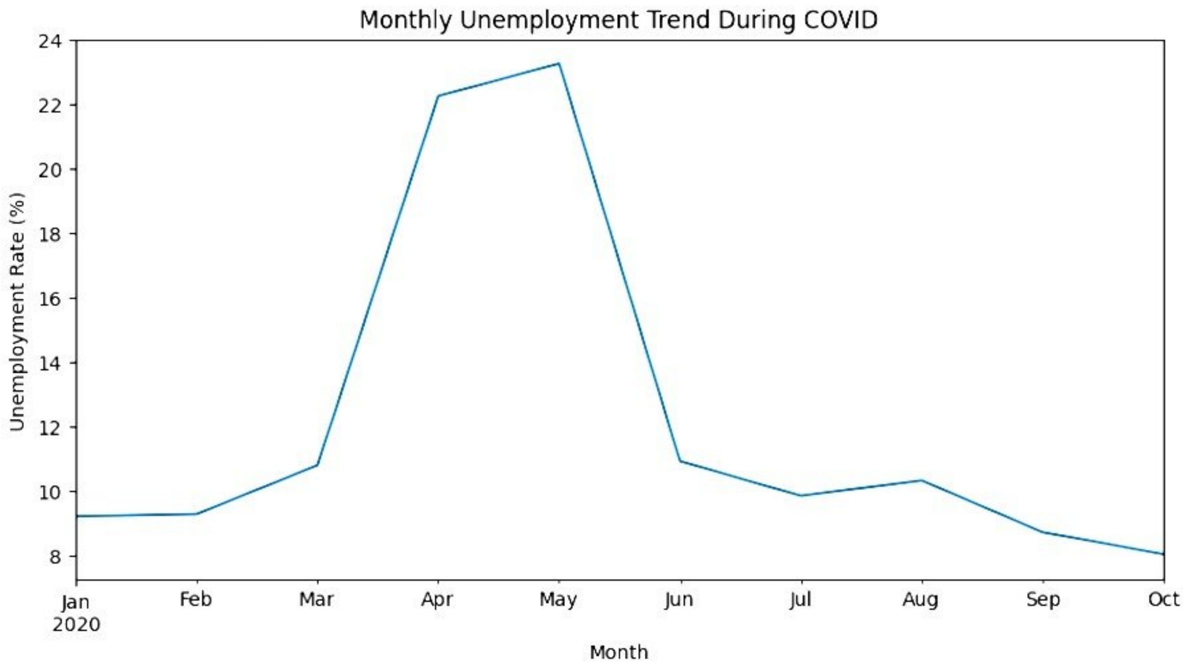
[9] 0s

```
df[' Date'] = pd.to_datetime(df[' Date'], dayfirst=True)

monthly = df.groupby(df[' Date'].dt.to_period('M'))[' Estimated Unemplo

monthly.plot(figsize=(10,5))

plt.title("Monthly Unemployment Trend During COVID")
plt.xlabel("Month")
plt.ylabel("Unemployment Rate (%)")
plt.show()
```



INSIGHT

1. Unemployment rates varied significantly across different states in India.
2. Some states showed consistently higher unemployment rates compared to others.
3. During the COVID-19 period in 2020, unemployment rates increased sharply.
4. Monthly unemployment trends indicate economic disruption caused by lockdowns and restrictions.
5. Data analysis and visualization helped identify important employment patterns and trends.